LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 2: Variables in Loops: Spirals and Growing Patterns

Module 1 | 45 minutes | Variables inside loops | Grades 3–5 · ages 8–10

Objective: Change a variable each time round a loop to make patterns that grow, like spirals, and to count.

Success criteria

- I can set a variable before a loop and change it inside.
- I can use a variable reporter inside a block.
- I can draw a growing spiral.

Key vocabulary

- **variable**: A named box that stores a value.
- **reporter**: A round block that gives a value, like a variable.
- **initialise**: Set a variable's starting value.
- **spiral**: A line that winds around, getting further from the middle.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Growing steps (5 min): a student walks a square path, taking 1 step, turn, 2 steps, turn, 3 steps... The class counts the growing variable aloud.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Square spiral: learners follow the steps.
14 min	**Perform**	Code Lab: square spiral, Trace the variable
9 min	**Probe**	Code Lab: counting robot, Debug it, activity
4 min	**Wrap-up**	Exit ticket: Write where the "set" block and the "change" block go in a spiral script, and why.

Unplugged warm-up

Growing steps (5 min): a student walks a square path, taking 1 step, turn, 2 steps, turn, 3 steps... The class counts the growing variable aloud.

Teaching the idea

Combining variables with loops lets each repetition be slightly different: the variable is set before the loop and changed inside it, and used (as a reporter) in a block. Spirals, counting and growing animations all follow this pattern.

Analogy: Like climbing stairs that get taller each step — same action, growing value.

Live demo: Draw a square spiral with the move block still set to 10 (bug): all lines the same. Ask what's missing; drag the length reporter in.

Questions to ask

Ask	Listen for
Where must the set block go?	Before the loop, or it resets every time.
What happens if the change is inside but the move uses a number?	Nothing grows; the variable is ignored.
How do we count backwards?	Change by −1.

Differentiation

Support
Pre-make the variable; provide the spiral script with one missing block for learners to add.

Stretch
Make a spiral whose turn angle is a variable that changes slowly (89, 90, 91...), or count in 5s.

Common misconceptions

Misconception	How to fix it
Setting the variable inside the loop	Set once before the loop; change inside.
Making a variable is enough	You must drop its reporter into a block to use its value.
change by sets the value	"change by" adds; "set to" replaces.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write where the “set” block and the “change” block go in a spiral script, and why.

Homework
Write the code to count down from 10 to 1, then say “Blast off!”.

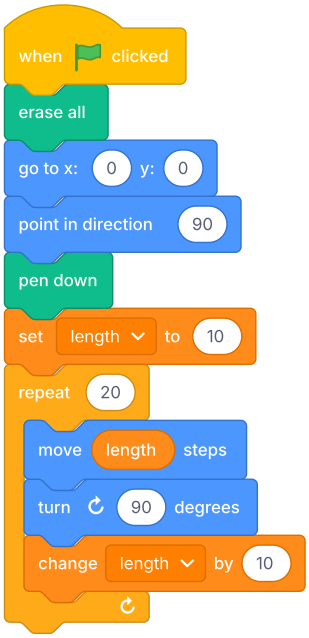
Answer key — Kitabash activities

1. Code Lab: square spiral (perform · code lab)

Goals checked: Moves by a variable · The variable grows · A growing spiral

One solution:

Pip



2. Trace the variable (perform · multiple choice)

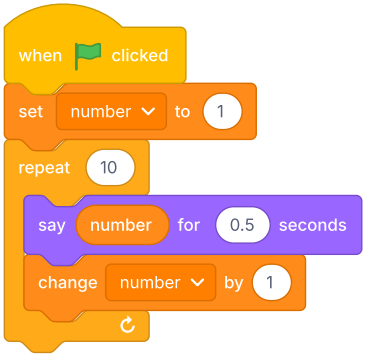
Answer: 25

3. Code Lab: counting robot (probe · code lab)

Goals checked: Counts 1, 5, ... 10 in order · Uses a loop · Says the variable

One solution:

Pip



4. Debug it (probe · multiple choice)

Answer: The move block uses the number 10 instead of the length variable

5. Explain your thinking (probe · free response)

Look for: variable, change, longer. **Example:** The variable length starts at 10. Every time the loop runs, it changes by 10, so the sprite moves a longer line each time. Longer and longer lines with turns make a spiral.

Answer key — worksheet

1. **Key words:** variable = A named box that stores a value. · reporter = A round block that gives a value, like a variable. · initialise = Set a variable's starting value. · spiral = A line that winds around, getting further from the middle.
2. **Plan your code:** Goals: Moves by a variable; The variable grows; A growing spiral.
3. **Trace the variable:** A — 10 + 5 + 5 + 5 = 25.
4. **Plan your code:** Goals: Counts 1, 5, … 10 in order; Uses a loop; Says the variable.
5. **Debug it:** A — The variable grows, but it is never used — move (length) steps is needed.

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